Fusion of Hyperspectral and Panchromatic Images Using Generative Adversarial Network and Image Segmentation

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***Abstract*— Hyperspectral (HS) image fusion aims at integrating a panchromatic (PAN) image and an HS image, featuring the fused image with the spatial quality of the former and the spectral diversity of the latter. The classic fusion algorithm generally includes three consecutive procedures that are upsampling, detail extraction, and detail injection. In this article, we propose an HS and PAN image fusion method based on generative adversarial network and local estimation of injection gain. Instead of upsampling the HS image by classical interpolation techniques, a generative adversarial super-resolution network (GASN) is designed to obtain the interpolated HS image in the fusion framework. GASN establishes a spectral-information-based discriminator to conduct adversarial learning with the generator, so as to preserve the spectral information of the low-resolution HS image. An image segmentation-based injection gain estimation (ISGE) algorithm is subsequently proposed for HS and PAN images fusion. The injection gain is estimated over image segments obtained by a binary partition tree approach to improve the fusion performance. The proposed GASN and ISGE are implemented into two credible global estimation pansharpening methods, and experimental results prove the performance improvement of the proposed method. The proposed method is also compared with existing state-of-the-art methods, and experiments on several public databases demonstrate that the proposed method is competitive or superior to the state-of-the-art fusion methods.**

***Index Terms*— Details injection, hyperspectral (HS) fusion, image segment, injection gains.**

I. Introduction

With the increasing demand for high-resolution data in many fields such as commerce, military, and agriculture, the study of hyperspectral (HS) fusion has become an important research topic. HS fusion is to fuse a high-spectral-diversity HS image with a high-spatial-resolution (HR) panchromatic (PAN) image to obtain a fused HS image with improved spatial resolution. The fused HS image should be as close as possible to the HS image acquired by a higher resolution sensor. In fact, due to the limitations of costs and physical constraints, it is impossible to achieve an image with both high spatial resolution and high spectral resolution by a single sensor. An HS fusion algorithm constitutes the possibility of achieving this goal. With the desirable resolution, the fused image can be of great value for subsequent applications, such as target recognition, change detection, and classification [1], [2].

The scientific community is increasingly interested in HS and PAN image fusion, and some HS and PAN image fusion algorithms have been proposed in recent years [3]. Most of these methods can generally be classified into the following categories: component substitution (CS), multiresolution analysis (MRA), Bayesian, matrix factorization, variational methods, tensor methods, and deep learning fusion approaches. Some research works have introduced deep learning-based methods into pansharpening tasks recently. Wei *et al.* [4] proposed a deep residual-based pansharpening network (DRPNN). Ma *et al.* [5] presented an unsupervised pansharpening method (PAN-GAN), which regards the pansharpening task as a multitask problem, aiming to preserve rich spectral information and maintain abundant spatial information simultaneously. Xu *et al.* [6] proposed a deep pansharpening network with two encoder–decoder networks to extract unique feature maps as the enhanced information representation. GTP-PNet [7] introduced the gradient transformation prior to the deep learning network, so as to reducing spectral distortion and spatial structure distortion. The variational fusion methods have attracted attention in recent years. Tian *et al.* [8] proposed a variational pansharpening method based on gradient sparse representation, which improves the spatial quality of the fused image. A variational pansharpening method that exploits cartoon–texture similarities was presented to preserve the global and local spatial details of the fused image well [9]. The tensor-based fusion methods define the HS image as a tensor and formulate the fusion problem as the estimation of a core tensor and dictionaries [10]. Dian *et al.* [11] proposed a low tensor-train rank (LTTR) method that designed an LTTR prior to learn the correlations among the spatial, spectral,

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and nonlocal modes. Qu *et al.* [12] proposed a structure tensor-based fusion (STF) method and introduced the structure tensor to extract spatial details of the PAN image. Bayesian comprises such classical algorithms as in [13]–[16]. Sparse representation-based Bayesian methods are adopted to enhance the spatial feature of HRMS by estimating the missing details [13], [17]. Among the works that rely on matrix factorization, a representative one is the coupled nonnegative matrix factorization (CNMF) [18], which unmixes data from two sources and generates a new image with both high spatial and spectral resolutions. Bayesian and matrix factorization methods will be able to provide the fused images with high quality by compromising computation complexity. However, they cannot meet the need for large-scale real-time system. The CS-based fusion approaches transform the HS image into a suitable space and then replace the spatial component of the HS image with the PAN image. The fusion approaches, such as intensity–hue–saturation (IHS) [19], principal component analysis (PCA) [20], [21], adaptive Gram–Schmidt (GS) [22], and adaptive GS (GSA) [23], are representatives of the CS class. Although these methods have high fidelity in rendering the spatial information and are quite robust to misregistration error and aliasing, they often generate a spectral distorted image because of the partial overlapping spectral range between the PAN and HS images. The MRA-based fusion approaches inject the spatial details obtained by multiscale decomposition of PAN images into the upsampled HS bands. MRA fusion class comprises such credible algorithms as smoothing filter-based intensity modulation (SFIM) [24] and the modulation transfer function—generalized Laplacian pyramid (GLP) method [25]. The MRA-based fusion techniques solve the problem of temporal coherence and spectral consistency, and unfortunately, their implementation is usually more complicated compared to CS methods due to the design of the spatial filter. Compared with other fusion categories, the fusion frameworks based on CS and MRA have the advantages of high efficiency and simple implementation.

In general, the CS- and MRA-based fusion methods contain three consecutive processes: 1) upsampling; 2) detail extraction; and 3) detail injection. The detail extraction refers to the estimation of the missing details in HS images [3], [26]. The CS-based fusion methods extract the details from the difference between the PAN image and the intensity component, which is a linear combination of the HS bands. The MRA-based fusion methods exploit the difference between the PAN image and its low-resolution version. Many existing CS- and MRA-based fusion methods are dedicated to maintaining the spectral information of HS images and estimating the details only from the high-frequency component of the PAN image. Detail injection refers to injecting the details estimated by the detail extraction step into the interpolated HS image, which heavily depends on the definition of the injection gains. Injection gains can be generally divided into two kinds: one is estimated globally and the other is spatially variant (or called the context-adaptive method). The context-adaptive methods usually provide the high-quality fused results by compromising computation complexity. The computational complexity of the global estimation methods is low enough to meet the requirements of the real-time system, but its performance is often poor especially when the contrast of the target in an image is large.

Since the low-spatial-resolution (LR) HS image and the HR PAN image have different spatial resolutions, the CS- and MRA-based fusion methods generally interpolate the LR-HS image to the scale of the HR PAN image and then inject the extracted details into the interpolated HS image through the injection gains defined in the detail injection step. Most upsampling methods in the CS and MRA classes are the basic nonlinear interpolation methods that utilize the values of neighboring pixels, such as bilinear and bicubic. However, these upsampling methods always have shortcomings that are edge blur and ringing effect. Recently, convolutional neural network (CNN)-based methods have been applied to the domain of interpolation and demonstrated superior performance. The outstanding advantage compared with traditional models is that CNNs have great potential to use their high nonlinearity to formulate the relationship between the interpolated HS image and the LR-HS image. Dong *et al.* [27] proposed the super-resolution CNN (SRCNN) to mapping the relationship between input and output images, which was the pioneering work to use CNN for image super-resolution (SR). Fast SRCNN substituted bicubic interpolation with the deconvolution layer to improve SRCNN [28]. Shi *et al.* [29] designed an efficient subpixel CNN to upsample single image and video, which proposed a subpixel convolutional layer to effectively improve the SR performance. These SR methods that are applied to a single RGB image can be extended to upsample the HS image. In addition, Hu *et al.* [30] used the CNN to characterize the spectral differences between LR-HS image and HR-HS image and exploited the spectral correlation for HS image SR. Wang *et al.* [31] applied the current band of the LR-HS image and its two adjacent bands to perform single-band SR and obtained the reconstructed HS image by the way of recurrent. Hu *et al.* [32] presented an intrafusion network (IFN) that utilized both the spatial and the spectral information for the SR process.

Although the traditional interpolation methods, such as nearest, bilinear, and bicubic can be very efficient, they always create serious reconstruction artifacts. The interpolation methods based on deep learning provided the superior reconstruction accuracy by using their nonlinearity. Some of these deep learning interpolation methods work like a black box [33] and do not define the physical interpretation. There are also some methods that only use the loss function to control spectral distortion, which may not maintain the spectral fidelity well. In order to preserve the spectral information of the LR-HS image, we propose a generative adversarial super-resolution network (GASN) to conduct upsampling on the LR-HS image in this article. GASN designs a spectral-information-based discriminator to play the adversarial game with the generator so that the spectral information of the interpolated HS image is consistent with that of the LR-HS image. In addition, the global injection gains estimation strategy has low complexity, but its performance is generally poor. The context-adaptive estimation method has a better performance at the expense of computational complexity. In order to

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overcome the abovementioned limitations, this article proposes an image segmentation-based injection gain estimation (ISGE) algorithm, which makes a tradeoff between computational complexity and fusion performance. The major contribution of the proposed ISGE is to estimate the injection gains over the segmented regions achieved via the binary partition tree (BPT). Experiments proved that the proposed method has a great potential to improve the fusion performance.

In summary, the main contributions of this article are summarized as follows.

1) We propose a region HS and PAN images fusion method based on generative adversarial network (GAN) and local estimation of injection gains, which maintains the spectral information of LR-HS image well and efficiently yields the HS image with high spatial and spectral resolution.
2) GASN is introduced to generate the interpolated HS image with high spectral fidelity. Instead of distinguishing the generated HS image from the HR-HS image, the discriminator of GASN spatially downsamples the generated HS image and distinguishes the spatially downsampled version of the generated HS image from the original LR-HS image.
3) An image segmentation-based region fusion method is introduced, in which the injection gain is estimated on each segmented region achieved via the BPT. The proposed gain estimation algorithm is applied to two credible global estimation pansharpening methods, and experimental results show that the proposed algorithm improves the performance of these methods.

The remainder of this article is organized as follows. Section II reports the proposed method. The experimental results are presented in Section III. Section IV draws the conclusion.

II. Proposed Method

Global estimation techniques will be able to provide a fast fusion method by compromising spectral distortion. Local estimation techniques always have a good fusion performance since the contrast of the PAN image remains unchanged in the fused image. However, the complexity of computation limits its application in some real-time systems. To overcome these drawbacks, we introduce a region fusion method in this article. BPT [34] is utilized to segment the image into multiple image regions, and the region merging criterion is defined by calculating a structure tensor matrix [35]. We estimate the injection gain separately on each image region and apply it to two representative global estimation-based pansharpening methods. In addition, the basic nonlinear interpolation methods, including bilinear and bicubic, are usually efficient, but they may create edge blur and ringing effect. In order to reduce edge blur and ringing effect, the proposed GASN replaces a traditional interpolation method to perform upsampling processing of HS image.

Therefore, we first introduce the proposed GASN and the ISGE algorithm and then apply them to two credible global estimation pansharpening methods.

A. Generative Adversarial Super-Resolution Network

In GAN, the generator and the discriminator are trained simultaneously in a competitive manner [36]. The GAN procedure learns reusable feature representations from a large amount of unlabeled data and generates the data that are closer to the real data. The generator of GAN is designed to generate the data from the input data, whereas the discriminator D aims to discriminate the real data from synthetically generated one. In this article, GASN is designed to upsample the LR-HS image to the same size as the PAN image. The generator G of GASN is responsible for learning to generate HR-HS image $\widetilde{\mathbf{H}}$ from the input LR-HS image $\mathbf{H}_{\mathrm{LR}}$. Specially, different from SRGAN that distinguishes the generated HR-HS image from real HR-HS image, an LR-HS discriminator is presented to distinguish the spatially downsampled version of the generated HR-HS image from the original LR-HS image, so as to make the spectral information of the generated HS image consistent with that of the LR-HS image.

1) Generator Network Architecture: The generator network of GASN, which is shown in Fig. 1, directly takes an LR-HS image $\mathbf{H}_{\mathrm{LR}}$ of size $w \times h \times l$ as an input to reduce computational overhead and estimates an HR-HS image $\widetilde{\mathbf{H}}$ from the input. The goal of the proposed GASN is to train a generating function G that can estimate the corresponding HR-HS image for the given LR-HS image. To achieve this goal, after a 3×3 convolutional layer, we design a feedforward CNN generator network embedded with an attention block and b residual blocks to extract spectral–spatial features. The attention mechanism is inspired by the attention model in human visual systems, which can magnify the contribution of important regions. For the image SR tasks, the main purpose is to make the network focus on a high-frequency component that contains more spatial features. Therefore, we introduce the channel attention module to help focus on "what" is meaningful features extracted by the former network layer and the spatial attention module as the complementary learning "where" is the informative part. The detailed structure of the attention block, which places channel and spatial attention modules in a sequential manner, is shown in Fig. 2(a). This process can be formulated as

$$\mathbf{F}_a = f_{\mathrm{conv}}(\mathbf{H}_{\mathrm{LR}}) + f_{sa}\{[f_{ca}(f_{\mathrm{conv}}(\mathbf{H}_{\mathrm{LR}}), \sigma_{s1})], \sigma_{s2}\} \quad (1)$$

where $\mathbf{F}_a$ denotes the output of attention block, σ_{s1} and σ_{s2} are the channel attention mask and the spatial attention mask followed sigmoid activation, respectively, and f_{ca} and f_{sa} denote the channelwise multiplication function and the elementwise multiplication function for the input and the corresponding mask, respectively. A deep residual network with b identical blocks is further integrated to learn high-level dimension features effectively. Each block is composed of two convolutional layers with 3×3 kernels, batch-normalization (BN) layers, and parametric rectified linear unit (PReLU) activation function between two convolutional layers. To generate the output HR-HS image of required spatial size, the subpixel convolutional layers are employed for more efficient upscaling and reconstruction, and the structure of upsample block is shown in Fig. 2(b).

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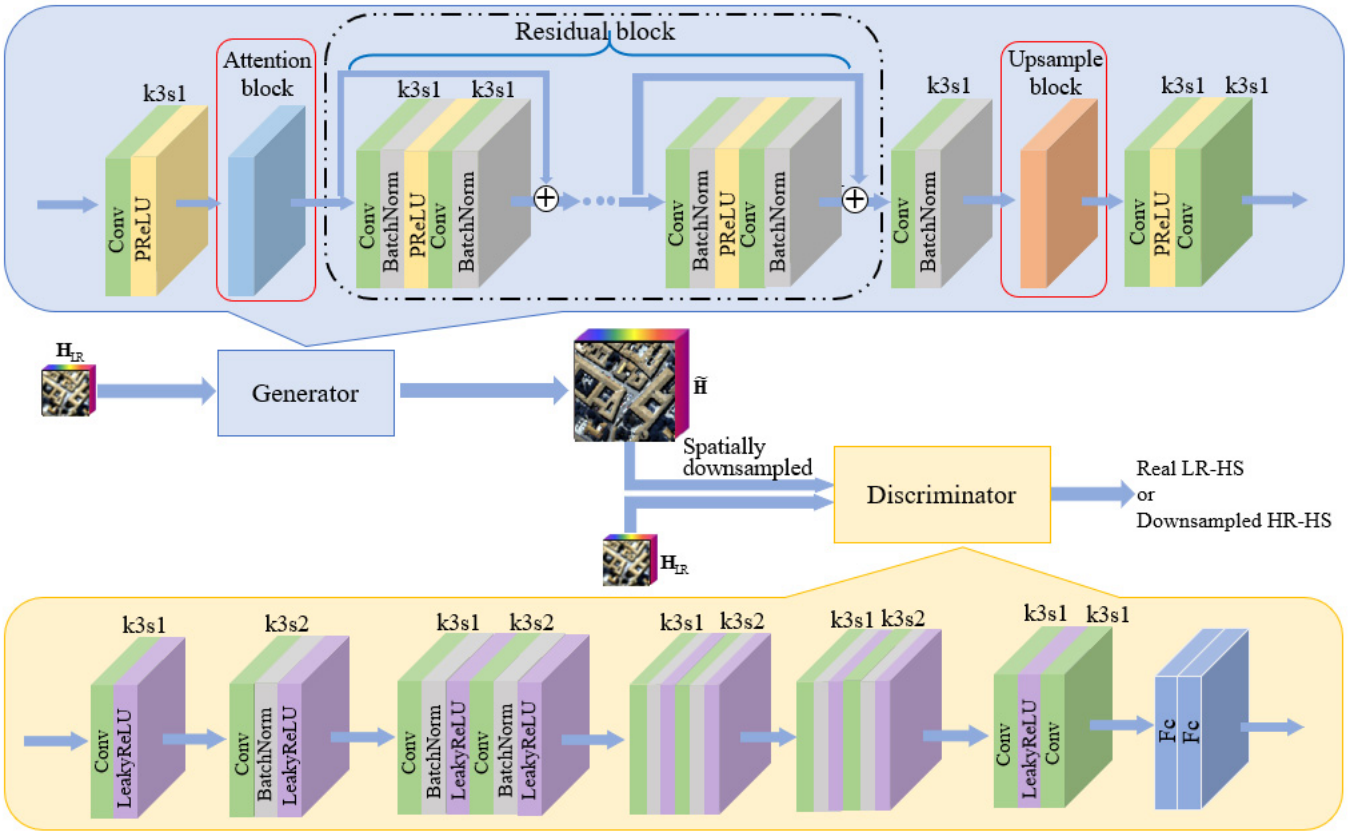


Fig. 1. Architecture of the proposed GASN including generator and discriminator networks. For each convolutional layer, the kernel size (k) and stride (s) are given.

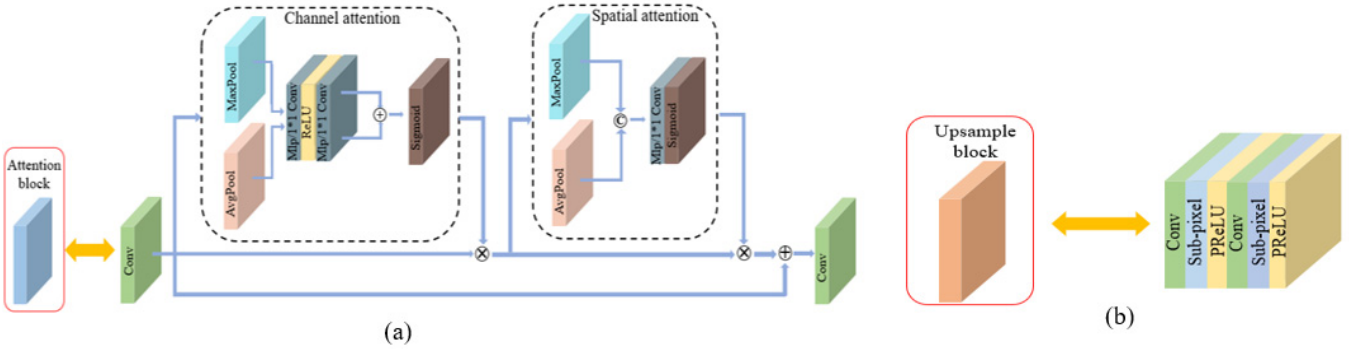


Fig. 2. Detailed structures of (a) attention block and (b) upsample block.

2) LR-HS Discriminator Network Architecture: Instead of distinguishing the generated HR-HS image from real HR-HS image, the proposed discriminator of GASN downsamples the generated HR-HS image $\widetilde{\mathbf{H}}$ to the same size as the LR-HS image and is learned to distinguish the synthetic LR-HS image $\widetilde{\mathbf{H}}_{\mathrm{LR}}$ from real LR-HS image $\mathbf{H}_{\mathrm{LR}}$ to avoid spectral distortion. The synthetic LR-HS image is obtained by spatially downsampling the HR-HS image, which can be represented as

$$\widetilde{\mathbf{H}}_{\mathrm{LR}} = \widetilde{\mathbf{H}}\mathbf{B}\mathbf{D} \tag{2}$$

where $\mathbf{B}$ denotes a convolution operator that is corresponding to the spectral response of the HS sensor expressed in the resolution of PAN image and $\mathbf{D}$ denotes the downsampling matrix. The generator of GASN is trained with the goal of fooling the discriminator that is utilized to correctly distinguish the downsampled image from the input LR image. A higher probability value given by the discriminator indicates that the spectral information of the generated HS image is highly consistent with that of the original LR-HS image. The structure of LR-HS discriminator network is shown at the bottom of Fig. 1, which utilizes leaky rectified linear unit (LeakyReLU) activation ($\alpha = 0.2$) to avoid sparse gradient during training. The architecture contains eight convolutional layers with the size of 3×3 kernels increasing from 128 to 1024, and the resolution of the image is reduced by a factor of 2 in every two layers. The resulting feature vectors

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are fed into two fully connected layers aiming at obtaining the probabilities for sample distinction.

3) Loss Function: Let $G(\mathbf{H}_{\mathrm{LR}};\beta_G)$ represent the generated HR-HS image obtained by the designed generator network with the trained parameters β_G. The construction of the generator loss function $l_G(\beta_G)$ is critical for pursuing an ideal result of the proposed GASN. We define the generator loss function by calculating the weighted sum of a pixel-based content loss and an adversarial loss, which can be represented as follows:

$$l_G(\beta_G) = l_{\mathrm{content}}(G(\mathbf{H}_{\mathrm{LR}};\beta_G),\mathbf{H}_{\mathrm{HR}}) + \gamma \cdot l_{\mathrm{adver}}(G(\mathbf{H}_{\mathrm{LR}};\beta_G),\mathbf{H}_{\mathrm{LR}}) \quad (3)$$

where γ is a tradeoff parameter that is set to 10^{-3} to balance the impacts of each term in (3) and $\mathbf{H}_{\mathrm{HR}}$ is the reference HR-HS image. The first term is the pixel-based content loss l_{content}, which is calculated by mean absolute error (MAE) between the generated HR-HS image $G(\mathbf{H}_{\mathrm{LR}};\beta_G)$ and the reference HR-HS image $\mathbf{H}_{\mathrm{HR}}$ and can be formulated as

$$l_{\mathrm{content}}(G(\mathbf{H}_{\mathrm{LR}};\beta_G),\mathbf{H}_{\mathrm{HR}}) = \frac{1}{WHC}\sum_{m=1}^{W}\sum_{n=1}^{H}\sum_{c=1}^{C}\left\|(\mathbf{H}_{\mathrm{HR}})_{m,n,c} - G(\mathbf{H}_{\mathrm{LR}};\beta_G)_{m,n,c}\right\|_1 \quad (4)$$

where W and H represent the width and height of the HR-HS image, respectively, C denotes the number of spectral channels, and $\|\|_1$ denotes the l_1-norm. l_1_loss has been an common optimization criterion for its simplicity in calculating and advantages in encouraging less blurring. The second term is the adversarial loss generated by the discriminator that encourages the generator to create satisfying image, which can be defined as follows:

$$l_{\mathrm{adver}}(G(\mathbf{H}_{\mathrm{LR}};\beta_G),\mathbf{H}_{\mathrm{LR}}) = -[D((G(\mathbf{H}_{\mathrm{LR}};\beta_G)\cdot\mathbf{BD});\beta_D)] \quad (5)$$

where $D((G(\mathbf{H}_{\mathrm{LR}};\beta_G)\cdot\mathbf{BD});\beta_D)$ is the probability that the spatially downsampled version of the generated HR-HS image is an LR-HS image and β_D is the trained parameters of discriminator. $-[D((G(\mathbf{H}_{\mathrm{LR}};\beta_G)\cdot\mathbf{BD});\beta_D)]$ is minimized instead of $[1-D((G(\mathbf{H}_{\mathrm{LR}};\beta_G)\cdot\mathbf{BD});\beta_D)]$ to obtain better gradient behavior.

As mentioned earlier, the proposed LR-HS discriminator distinguishes the spatially downsampled version of the generated HR-HS image from real LR-HS image $\mathbf{H}_{\mathrm{LR}}$. The traditional GANs have the problems of mode collapse and vanishing, and the Wasserstein GAN (WGAN) [37] that uses weight clipping to improve the stability has the problems of optimization difficulties. In order to solve these problems, WGAN-gradient penalty (WGAN-GP) [38] that uses gradient penalty is adopted to optimize the LR-HS discriminator. The loss function of LR-HS discriminator is defined as

$$l_D(D_G) = [D(G(\mathbf{H}_{\mathrm{LR}};\beta_G)\cdot\mathbf{BD});\beta_D] - [D(\mathbf{H}_{\mathrm{LR}};\beta_D)] + \varepsilon[\|\nabla_z D(z;\beta_D)\|_2 - 1] \quad (6)$$

where ε is a gradient penalty coefficient that is set to 10 based on a variety of comparative experiments and z denotes random samples subjecting to uniform distribution, which is defined as

$$z = \rho\cdot\mathbf{H}_{\mathrm{LR}} + (1-\rho)\cdot[G(\mathbf{H}_{\mathrm{LR}};\beta_G)\cdot\mathbf{BD}],\ \ \rho\sim\mathrm{uniform}[0,1]. \quad (7)$$

B. Image Segmentation-Based Injection Gain Estimation

BPT [34] provides an effective representation tool for image segmentation. In BPT, leaf nodes represent the initial fine image region, and nonleaf nodes represent the new region obtained by merging two similar regions. The construction of a BPT is an iterative process that merges two most similar regions. In each iteration, we first use a merging criterion to find the most similar pair of regions. Then, the two most similar regions are merged to get a new region, and the nonleaf node is used to represent this new region. Finally, the similarity between the new region and its neighbors is estimated for the next iteration. It can be seen from the iterative process that the region model and the merge criterion are important factors in constructing BPT.

The gradients of the edges and corner points of the image change greatly, whereas the gradients of the flat areas of the image are small. The gradient can capture information on edges, textures, corners, and more. The structure tensor [35] is a matrix derived from the gradient of the function. It contains neighborhood structure features of image and preserves the details of images edge. The structure tensor describes the spatial information of the image by the gradient information of the pixels in a specified neighborhood. It is often used to distinguish corners, edges, and flat regions in an image.

Most of the CS- and MRA-based fusion algorithms use the PAN image as the ideal detail image. A good fused result should preserve the spatial detail contrast of the PAN image. Inspired by the above analysis, it is expected that pixels with similar gradient variation in PAN image require equal injection gain. Therefore, it is really a good way to define a region model using the structural tensor of the PAN image. Specifically, the structure tensor of the ith pixel of a PAN image $\widehat{\mathbf{T}}_i$ can be expressed as

$$\widehat{\mathbf{T}}_i = \begin{bmatrix}\mathbf{P}_{x,i}^2 & \mathbf{P}_{x,i}\mathbf{P}_{y,i}\\ \mathbf{P}_{x,i}\mathbf{P}_{y,i} & \mathbf{P}_{y,i}^2\end{bmatrix} \quad (8)$$

where $\mathbf{P}_{x,i}$ and $\mathbf{P}_{y,i}$ are, respectively, the horizontal and vertical gradients of the ith pixel of PAN image. By smoothing the structure tensor $\widehat{\mathbf{T}}_i$ with filtering technology, the structure tensor $\mathbf{T}_i$ containing the structure information of the neighborhood of pixel i can be obtained

$$\mathbf{T}_i = \widehat{\mathbf{T}}_i * F_g = \begin{bmatrix}g_{1,1} & g_{1,2}\\ g_{2,1} & g_{2,2}\end{bmatrix} \quad (9)$$

where F_g refers to the Gaussian function and $*$ represents the convolution operator. Since $\mathbf{T}_i$ is a semidefinite matrix, (9) can be expressed as

$$\mathbf{T}_i = \begin{bmatrix}\eta_{1,i} & \eta_{2,i}\end{bmatrix}\begin{bmatrix}\lambda_{1,i} & 0\\ 0 & \lambda_{2,i}\end{bmatrix}\begin{bmatrix}\eta_{1,i} & \eta_{2,i}\end{bmatrix} \quad (10)$$

where $\lambda_{1,i}$ and $\lambda_{2,i}$ are the two nonnegative eigenvalues of the structure tensor $\mathbf{T}_i$ and $\eta_{1,i}$ and $\eta_{2,i}$ are the corresponding

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eigenvectors. $\lambda_{1,i}$ and $\lambda_{2,i}$ can be solved by the following equation:

$$\lambda_{1,i} = \frac{1}{2}[g_{1,1} + g_{2,2} + \sqrt{(g_{1,1} - g_{2,2})^2 + 4g_{1,2}^2}] \quad (11)$$

$$\lambda_{2,i} = \frac{1}{2}[g_{1,1} + g_{2,2} - \sqrt{(g_{1,1} - g_{2,2})^2 + 4g_{1,2}^2}]. \quad (12)$$

When $\lambda_{1,i} \approx \lambda_{2,i} \approx 0$, pixel i is located in the smooth region of the PAN image with the greatest probability. When $\lambda_{1,i} \gg \lambda_{2,i} \approx 0$, pixel i is most likely to be located in the edge region of the PAN image. When $\lambda_{1,i} \approx \lambda_{2,i} \gg 0$, pixel i is most likely a pixel on a corner of the PAN image. Through a lot of experiments, we found that for most pixels in the image that the smaller eigenvalues are usually very small. This is because there are far more edges pixels than corner pixels for the object in an image. Based on the above analysis, we construct a region model $\mathcal{M}_R$ based on the maximum eigenvalue of the structure tensor

$$\mathcal{M}_R = \frac{1}{|R|} \sum_{i \in R} \lambda_i \quad (13)$$

where $|R|$ represents the number of pixels in region R and λ_i represents the larger eigenvalue of the structural tensor at pixel i in the region R. In order to calculate the similarity between regions, the region merging criterion is defined as

$$\mathcal{C}(\mathcal{M}_{R_i}, \mathcal{M}_{R_j}) = \min |\mathcal{M}_{R_i} - \mathcal{M}_{R_j}|. \quad (14)$$

Then, the initial image segment can be obtained by the region model and merging criterion.

The image segmentation is achieved by iteratively pruning the BPT. Two regions are merged according to the merging criterion in each cut. The goal of the ISGE algorithm is to propose an injection gain estimation algorithm that improves the performance of the global estimation pansharpening methods and balances fusion performance with computational complexity. The number of segmented regions in the segmentation map determines the computational complexity. We control the computational complexity of the proposed ISGE method by fixing the number of segmented regions. The global injection gain methods estimate a single injection coefficient for each whole spectral band. ISGE estimates the injection coefficients on each segmented region.

C. GS and GLP Using GASN and ISGE

In this section, we apply the proposed GASN and ISGE algorithms to two credible global estimation pansharpening methods that are the GS and GLP methods. The GS and GLP algorithms with high credibility belong to the CS and the MRA class, respectively. As mentioned in Section I, the CS- and MRA-based fusion methods contain three consecutive processes that are upsampling, detail extraction, and detail injection. The upsampling step of the GS and GLP algorithms generally utilizes the traditional basic interpolation methods, and we replace it with the proposed GASN in the upsampling step to improve edge blur and ringing effect.

For GS, the detail image is a difference map between the PAN image and the intensity component of HS image, and for GLP, the detail image is a difference map of the PAN image and its low-resolution version. The formula is expressed as

$$\mathbf{D} = \mathbf{P} - \mathbf{L} \quad (15)$$

in which, for GS, $\mathbf{L}$ can be calculated by

$$\mathbf{L} = \sum_{j=1}^{J} \widehat{\omega}_j \widetilde{\mathbf{H}}_j \quad (16)$$

where the expressions of the weights $\{\widehat{\omega}_j\}_{j=1,\ldots,J}$ are taken as $1/J$ and J is the bands number in HS image. For GLP, $\mathbf{L}$ is calculated as

$$\mathbf{L} = \mathbf{P} * f_h \quad (17)$$

where f_h is a linear time-invariant filter.

The ISGE algorithm is applied to the GS and GLP methods to estimate the injection gains. The GS and GLP methods share the same definition of the injection gains. For the jth band, the injection gain of the ith pixel in the R region $g_i^{(R)}$ can be expressed as

$$\mathbf{g}_{j,i}^{(R)} = \frac{\operatorname{cov}\left(R_i^{\widetilde{\mathbf{H}}_j}, R_i^{\mathbf{L}}\right)}{\operatorname{var}\left(R_i^{\mathbf{L}}\right)} \quad (18)$$

where (R) denotes the segment region R and $R_i^{\widetilde{\mathbf{H}}_j}$ and $R_i^{\mathbf{L}}$ are, respectively, the R region in the images $\widetilde{\mathbf{H}}_j$ and $\mathbf{L}$ that contains pixel i. Finally, for the jth band, the fused HR-HS image $\widehat{\mathbf{H}}$ can be expressed by the following prescriptive formula:

$$\widehat{\mathbf{H}}_j = \widetilde{\mathbf{H}}_j + \mathbf{g}_j \circ \mathbf{D}_j \quad (19)$$

where $j = 1, 2, \ldots, J$, and $\circ$ represents the element-by-element multiplication operation.

III. Experiment Results and Analysis

A. Databases and Experimental Setup

To evaluate the fusion performance of the proposed method, a series of experiments was conducted on the Cave database and Pavia Center database. The Cave database acquired by a generalized assorted pixel camera captures a wide variety of real-world materials and objects. The dimension of the HS images is 512×512 pixels and this database contains 32 different scenes. An HS image comprises 31 spectral bands and covers a spectral range of 400–700 nm. According to Wald's protocol [39], the available high-resolution HS image is used as a reference image. A semisynthesized LR-HS image is obtained by downsampling and blurring the reference image, and the visible bands of the reference image are averaged to generate a PAN image. The spatial resolution of the LR-HS image is 128×128 and that of the PAN image is 512×512. We used 22 images randomly selected from the Cave database to train our proposed GASN model and the rest images were used for testing. Due to the lack of training data, the LR-HS image, PAN image, and reference HS image are divided into 16 patches without overlapping areas, the scales of which are $32 \times 32 \times 31$, 128×128, and $128 \times 128 \times 31$, respectively.

The Pavia Center database presents an urban area of Pavia, Italy. It was obtained by the ROSIS-3 optical

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airborne sensor. The HR-HS images cover the spectral range from 430 to 860 nm with 115 bands. After removing 12 noise bands and water absorption bands with low signal-to-noise ratio (SNR), the remaining 103 bands are used for experiments. The spatial dimension of the HS image is 1096×1096. However, because it is necessary to discard some samples that do not contain any information, the size of the HS image used in the experiment is 960×640. We divide the image into 12 nonoverlapping small patches so that all training and testing patches are with the size of $160 \times 320 \times 102$. We randomly select nine patches to generate the training set, and the remaining three patches are used for testing. In order to obtain sufficient training data, we further slice training samples into 21 patches whose size is set as $160 \times 160 \times 102$. Based on Wald's protocol, the HS and PAN images are degraded to a lower resolution with the spatial size of 40×40 and 160×160, respectively.

The proposed method is implemented with the open-source PyTorch framework and trained on NVIDIA GTX 2080Ti GPUs. For each dataset, the GASN is trained in batches of which size is 8. The Adam modification [40] is employed to optimize the parameters of network, and the learning rate is initialized to 2×10^{-4}, which is decreased by a factor of 10 per 256 epochs.

B. Compared Methods and Quality Assessment

To verify the effectiveness of the proposed model, the GS [22] and GLP [25] methods are two credible global estimation pansharpening methods and have been employed for comparison. The GS and GLP methods are two effective fusion approaches, belonging to the CS and the MRA class. The proposed GASN and ISGE algorithms are applied to GS and GLP. Several state-of-the-art pansharpening methods, including CNMF [18], Bayesian [16], STF [12], GFPCA [41], and DRPNN [4], are also used for comparison.

On the one hand, the focus of the experiment is a comparison of the fusion performance of GS and GLP methods, in which the proposed algorithms are applied to them for verifying the fusion performance. The proposed GASN algorithm, which refers to the proposed SR algorithm, is utilized to replace the traditional basic interpolation methods in the upsampling step of the GS and GLP algorithms and is named GS_GASN and GLP_GASN, respectively. The proposed ISGE algorithm that estimates the injection coefficients on each segmented region replaces the global injection gains estimation of GS and GLP and is named GS_ISGE and GLP_ISGE, respectively. The GASN and ISGE algorithms that are applied to GS and GLP are named GS_pro, and GLP_pro, respectively. On the other hand, for further showing the advantages of the proposed method, the GFPCA is a hybrid method that effectively combines the advantages of CS and MRA. The Bayesian Sparse and CNMF, which belong to the Bayesian class and the matrix factorization class, respectively, are recently proposed and widely accepted model-based methods and exhibit high quality in both spatial and spectral aspects. The STF method that proposes an effective way to balance spatial distortion and spectral fidelity is selected for comparison. The DRPNN method is a state-of-the-art deep learning-based pansharpening algorithm and is also conducted in comparative experiments.

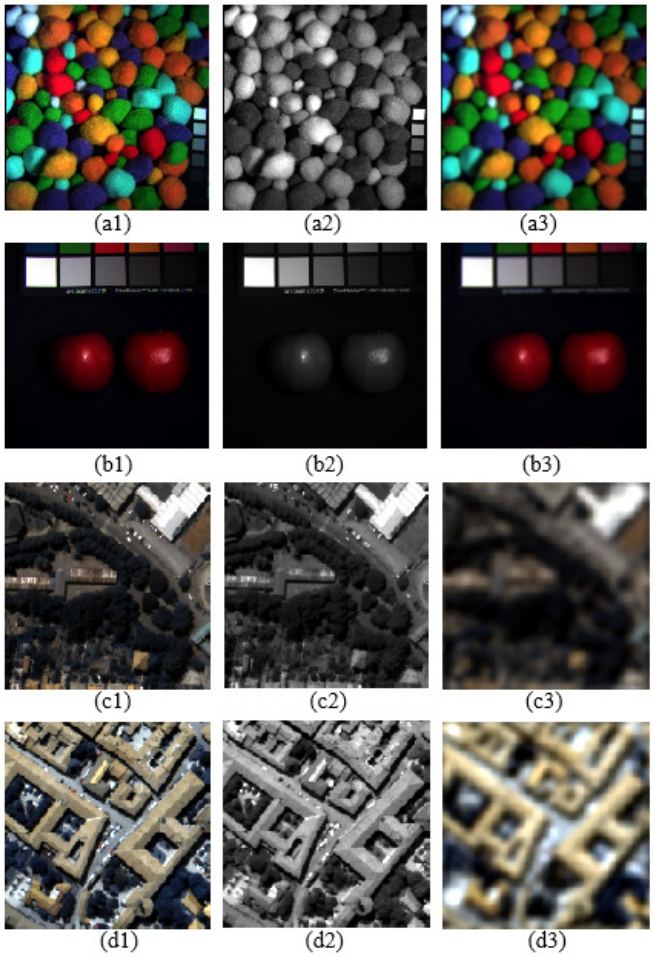


Fig. 3. Experimental databases. (a1)–(a3) One random test image from the Cave database. (b1)–(b3) Another random test image from the Cave database. (c1)–(c3) One random test image from the Pavia Center database. (d1)–(d3) Another random test image from the Pavia Center database. The first, second, and third columns show the reference HS images, simulated PAN image, and simulated HS image of each data, respectively.

To objectively evaluate the quality of the fused image, four widely accepted indexes are adopted, including the correlation coefficient (CC), spectral angle mapper (SAM) [42], root-mean-squared error (RMSE), and erreur relative globale adimensionnelle de synthèse (ERGAS) [43]. CC is a spatial metric that evaluates spatial distortion by averaging the CC values of all bands, and its optimal value is 1. SAM is a spectral index that characterizes the fidelity of the spectral by measuring the similarity of the spectral vectors of the corresponding pixels on the fused image and the reference image. The closer to 0 the SAM is, the higher spectral similarity between the fused image and reference one is. RMSE and ERGAS are both global indexes measuring the overall quality of the pansharpened image and are expected to close to 0.

C. Experimental Results

Two different databases were experimentally analyzed in this section. Fig. 3(a1)–(a3) and (b1)–(b3) shows two

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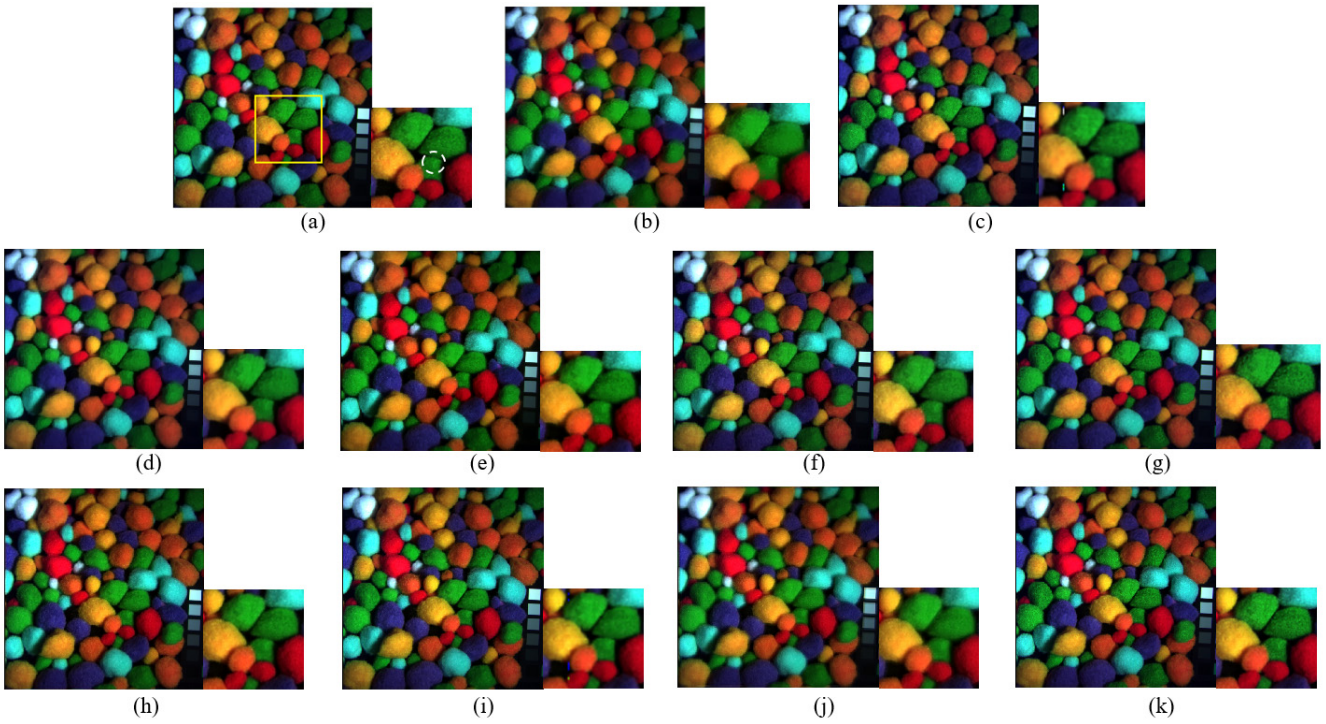


Fig. 4. Visual results of one HS image from the Cave dataset. (a) Reference HS image. (b) GFPCA. (c) STF. (d) Bayesian. (e) CNMF. (f) DRPNN. (g) LTTR. (h) GS. (i) GS_pro. (j) GLP. (k) GLP_pro.

test images randomly selected in the Cave database, and Fig. 3(c1)–(c3) and (d1)–(d3) shows two test images randomly acquired from the Pavia Center database. The first column shows the reference HR-HS images, and the second and third columns show the simulated HR-PAN and LR-HS images, respectively.

1) Experimental Results of Cave Database: The visual results of different pansharpening methods on two different databases are shown in Figs. 4 and 6. The proposed GASN and ISGE algorithms that can improve the fusion performance have been tested in both GS and GLP methods. By observing Fig. 4, it can be seen that the pansharpened images generated by our GS_pro and GLP_pro methods look much more similar to the reference image, with no obvious spectral distortion and edge blurring. As for the pansharpened images yielded by the STF and DRPNN methods, we can see that the spectral information tends to change in the blue and yellow areas, which testifies the poor spectral fidelity. Fig. 4 also shows a local square area of each product, and this area is enlarged for a clear spatial detail comparison. Insufficient spatial details exist in the fusion results of the GFPCA, Bayesian, CNMF, LTTR, and GLP methods, especially at the edges of the balls, although they do well in spectral preservation. The main reason for the burring effect may be that the spatial details injected into HS image are not enough. Some obvious artifacts can be seen in the enlarged subregion of the image generated by the STF method, which is caused by the injection of much more spatial information. To better compare the performance of different algorithms, we show the difference images between the reference HR-HS image and the fused HR-HS images generated by each competing method in Fig. 5, where dark blue represents smaller differences. It can be obviously seen that the proposed GS_pro and GLP_pro methods achieve the smaller difference values than the original GS and GLP methods, which gives the evidence of the performance improvement of the proposed method.

Since some slight differences could not be observed from the subjective results, we use four widely used indexes for qualitative evaluation, and the quantitative results of different HS pansharpening methods on the Cave database are shown in Table I. The GFPCA method performs worse than other methods for almost all the indexes. The SAM of the CNMF method is higher than most competing methods for the Cave database, demonstrating the poor performance in spectral preservation. The optimal value for each index is shown in the table, and the best results for GS and GLP are shown in bold. From Table I, it can be seen that the four indexes provided by the GASN are better than the original ones for both GS and GLP methods. Similar to the quantitative results of the GASN, the ISGE also achieves better scores than the original ones in terms of four assessment indexes. It is easy to notice that the GS_pro method gets the best results in the Cave database, which verifies the effectiveness of the GASN and ISGE in improving the pansharpening performance.

2) Experimental Results of Pavia Center Database: Fig. 6 shows the fused results produced by different methods over the Pavia Center database. It can be seen that the results of GFPCA, Bayesian, LTTR, CNMF, and GS methods show different degrees of color difference relative to the reference image, which is mainly caused by spectral distortion.

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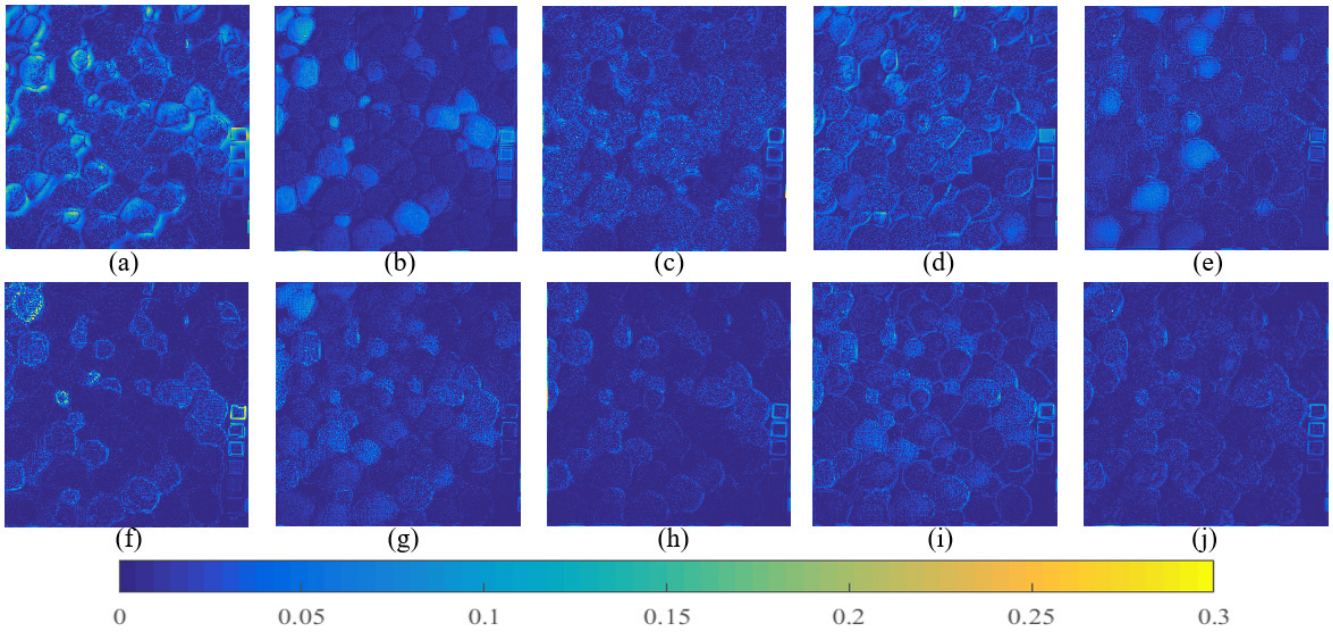


Fig. 5. Error images of Cave database. (a) GFPCA. (b) STF. (c) Bayesian. (d) CNMF. (e) DRPNN. (f) LTTR. (g) GS. (h) GS_pro. (i) GLP. (j) GLP_pro.

TABLE I
AVERAGE QUANTITATIVE ASSESSMENT FOR THE EXPERIMENTAL DATABASES

Database		Cave				Pavia Center			
Index		CC	SAM	RMSE	ERGAS	CC	SAM	RMSE	ERGAS
Reference		1	0	0	0	1	0	0	0
GFPCA		0.9874	4.9250	0.0238	5.3594	0.8519	9.0463	0.0552	8.2134
STF		0.9923	4.7523	0.0211	4.0125	0.9091	6.9372	0.0398	5.8821
Bayesian		0.9942	5.5665	0.0144	3.6403	0.9129	8.5743	0.0406	6.3339
CNMF		0.9930	7.0365	0.0166	3.8170	0.8964	6.9006	0.0426	6.4953
LTTR		0.9943	4.7017	0.0139	2.9469	0.8868	9.8954	0.0538	6.9930
DRPNN		0.9948	4.7588	0.0138	2.8669	0.8896	9.7898	0.0433	7.5057
GS	original	0.9946	4.9504	0.0148	3.4234	0.9071	7.2350	0.0425	6.2060
	GASN	**0.9971**	3.9251	0.0123	2.4624	**0.9273**	6.9489	0.0388	5.6223
	ISGE	0.9951	4.0375	0.0129	2.5702	0.9234	6.9938	0.0401	5.6297
	pro	0.9965	**3.9246**	**0.0121**	**2.3153**	0.9231	**6.7432**	**0.0372**	**5.4402**
GLP	original	0.9880	4.3460	0.0156	3.7435	0.9093	6.9245	0.0398	5.8113
	GASN	0.9923	3.9112	**0.0140**	2.9435	0.9268	6.5481	0.0353	5.5232
	ISGE	0.9924	3.8421	0.0142	2.9712	**0.9379**	5.3963	0.0330	4.8550
	pro	**0.9948**	**3.7263**	0.0142	**2.8987**	0.9313	**5.3205**	**0.0323**	**4.6872**

Although the DRPNN method achieves promising performance in reducing spectral distortion, the fuzzy roof edges indicate that it cannot well retain the spatial details of the HS image. Meanwhile, the texture features of the pansharpened images obtained by the Bayesian and CNMF methods cannot be well maintained, which is caused by insufficient spatial information injected into the LR-HS image. The fidelity of the GS_pro and GLP_pro methods in reconstructing the edges of the HS image is clear with respect to the comparing methods, including the original GS and GLP methods. Compared with the original GS and GLP methods, the HS images reconstructed by the GS_pro and GLP_pro methods show better spectral performance and clearer edge texture, indicating the effectiveness of the proposed GASN and ISGE algorithms in improving spectral and spatial fidelity. Furthermore, the results of the GS_pro and GLP_pro methods show the smallest difference with the reference image in terms of spectral fidelity, particularly the roofs in the top-left corner of Fig. 6(h) and (j). In order to further compare the fusion performance, the error images (absolute values) of different algorithms for the Pavia Center database are shown in Fig. 7. From Fig. 7, the error images generated by the proposed GS_pro and GLP_pro methods show the smaller difference values than that of other comparative methods, which testifies the superior fusion

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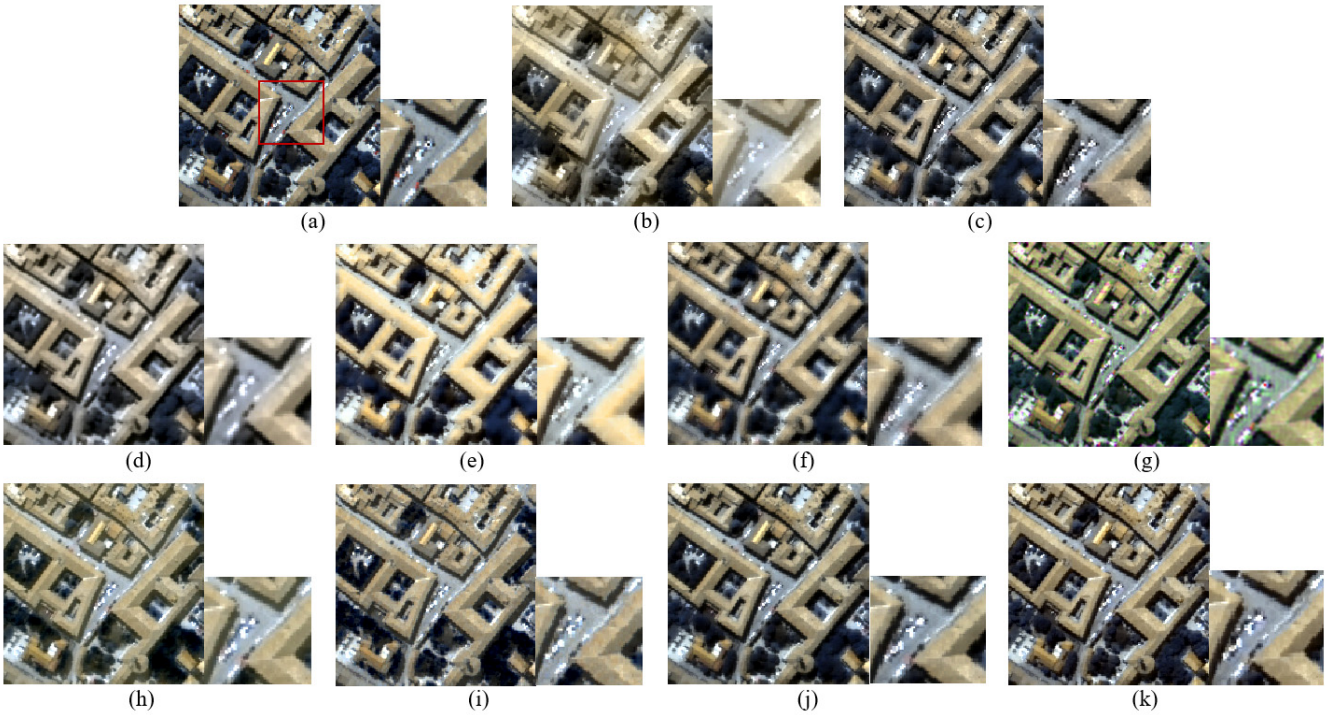


Fig. 6. Visual results of one HS image from the Pavia Center database. (a) Reference HS image. (b) GFPCA. (c) STF. (d) Bayesian. (e) CNMF. (f) DRPNN. (g) LTTR. (h) GS. (i) GS_pro. (j) GLP. (k) GLP_pro.

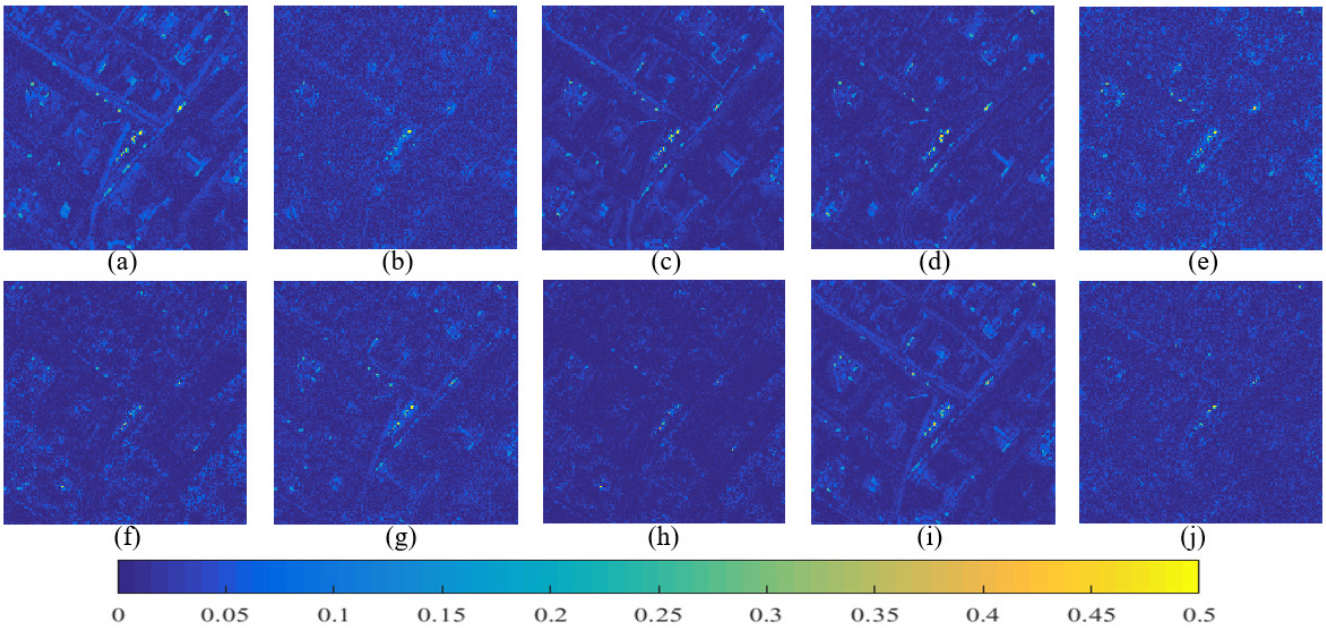


Fig. 7. Error images of Pavia Center database. (a) GFPCA. (b) STF. (c) Bayesian. (d) CNMF. (e) DRPNN. (f) LTTR. (g) GS. (h) GS_pro. (i) GLP. (j) GLP_pro.

performance of the proposed method. The proposed GS_pro and GLP_pro methods also achieve the smaller differences than the original GS and GLP methods, which proves that the proposed method can improve the fusion performance.

The results of the quantitative assessment achieved by the proposed methods and the competing methods for the Pavia Center database are presented in Table I. It can be seen that the GS_GASN and GLP_GASN methods are better than the original GS and GLP methods, which confirms that the GASN reduces the effects of spatial and spectral distortion in the traditional interpolation. Moreover, the GS and GLP methods using the ISGE strategy also achieve better pansharpening performance than the original ones. It is clear that the GS_pro and GLP_pro methods get the best values in terms of most

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TABLE II
AVERAGE COMPETING TIME OF DIFFERENT ALGORITHMS

Method	GFPCA	STF	Bayesian	CNMF	LTTR	DRPNN	GS	GASN	GS_pro	GLP	GLP_pro
Time (s)	0.543	5.992	46.928	1.157	27.104	0.036	0.304	0.015	45.078	2.023	45.289

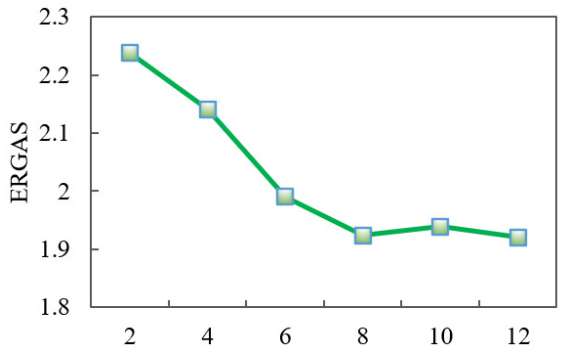


Fig. 8. Fusion results of the proposed GASN method with a different number of residual blocks on one Cave image shown in Fig. 4.

indexes compared with other three GS- and GLP-based methods, demonstrating the superiorities of GASN and ISEG in preserving spectral and spatial information. Serious spectral distortion is illustrated by the SAM indexes for GFPCA, Bayesian, and DRPNN methods. The CC value of the CNMF method is significantly lower than that of other methods except for GFPCA, illustrating that the spatial information is not well improved. The quantitative values are consistent with the visual results, which proves the superior ability of the GS_pro and GLP_pro methods.

The computing time of each competing method is also compared, and all the experiments are implemented in MATLAB (R2015b) with Intel Core i5-7300HQ CPU@2.50-GHz CPU and NVIDIA GTX 2080Ti GPU. The computational costs of different competitors on the Pavia Center database are given in Table II, in which the results are recorded in seconds. From Table II, the GFPCA, GS, and GSA methods are very efficient, but the pansharpened HR-HS images generated by these methods are unsatisfactory. It can be observed that the proposed GASN achieves high computation efficacy. The Bayesian, LTTR, GS_pro, and GLP_pro methods are more time-consuming. Although the proposed GS_pro and GLP_pro are not advantageous in terms of calculation time, the GS_pro and GLP_pro methods obtain better fusion performance than other methods, especially for the Pavia Center database. In particular, the proposed GASN and ISGE that are applied to the GS and GLP methods effectively improve the fusion performance of GS and GLP.

3) Experiments on Residual Blocks: In the generator network of GASN, b residual blocks are utilized to extract spectral–spatial features of the HS image. To analyze the fusion performance of different numbers of residual blocks, we conduct experiments on the Cave1 image, and the difference is the number of residual blocks that are set to 2, 4, 6, 8, 10, and 12. The results are evaluated with ERGAS since the global index is capable of measuring the overall quality of the fusion image. Fig. 7 shows the fusion performance of the proposed GASN method with a different number of residual blocks. As shown in Fig. 7, when the number of residual blocks increases, the performance improves. When the number exceeds 8, the ERGAS tends to be stable for the proposed GASN, indicating that the deeper structures do not always result in better performance. Therefore, we choose the network with eight residual blocks in this article to achieve a tradeoff between complexity and performance.

IV. CONCLUSION

In this article, we have introduced a region HS and PAN images fusion method based on GAN and local estimation of injection gains, in which the injection gains are calculated on the segmented regions. All pixels divided in the same region share the same injection gain. The injection gain estimation ISGE scheme proposed in this article can effectively balance the oversharpening and blurring of fused images. Instead of employing bicubic interpolation to conduct upsampling on the LR-HS image, GASN is introduced to generate the interpolated HS image. GASN that establishes an LR-HS discriminator network tends to maintain spectral details better and is beneficial to the following fusion process. The proposed GASN and ISGE algorithms are applied to two credible global estimation pansharpening methods that are the GS and GLP methods. The empirical results show that the proposed GASN and ISGE algorithms improve the performance of the GS and GLP methods, and the GS_pro and GLP_pro methods obtain the superior performance over competitors, both quantitatively and perceptually.

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